Microbiology: Emerging Infections

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Session Objectives

- Define emerging infections.
- Explore emerging viral and bacterial pathogens, their properties, pathogenesis, etc.
- Discuss the causes that lead to the emergence of these diseases and efforts to combat that.
- Discuss the challenges emerging infectious diseases present to the medical community.

Source: Course Syllabus

Emerging Infectious Diseases

“Emerging infectious diseases are infections that have recently appeared within a population or those whose incidence or geographic range is rapidly increasing or threatens to increase in the near future.”

https://www.bcm.edu/departments/molecular-virology-and-microbiology/emerging-infections-a

Emerging Infectious Diseases

- Emerging infections can be caused by:
  - Previously undetected or unknown infectious agents
  - Known agents that have spread to new geographic locations/populations
  - Previously known agents whose role in specific diseases has previously gone unrecognized
  - Re-emergence of agents whose incidence of disease had significantly declined in the past, but whose incidence of disease has reappeared (re-emerging)
  - Most Zoonotic

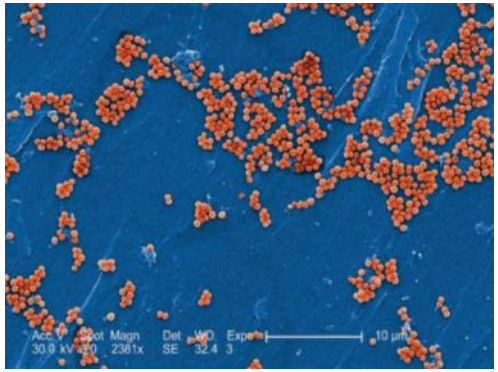
https://www.bcm.edu/departments/molecular-virology-and-microbiology/emerging-infections-and-biodefens

EID List (WHO/CDC)

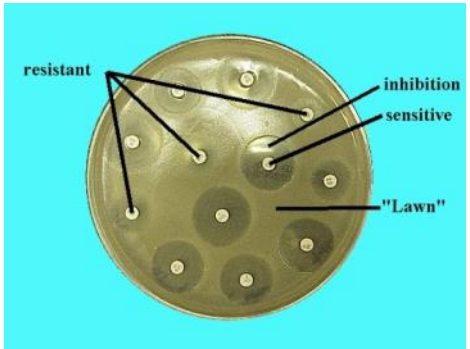
- Viral:
  - **Mumps**
  - **Measles**
  - **Influenza** (avian & swine)
  - HIV/AIDS
  - **Dengue fever**
  - **Ebola virus disease**, **Marburg** & Lassa fever
  - **MERS, SARS-CoV-2** coronavirus diseases
  - **Hantavirus pulmonary syndrome**
  - Crimean Congo hemorrhagic fever
  - Nipah & Rift Valley fever
  - Chikungunya virus disease
  - Zika virus disease
- Bacterial:
  - Antibiotic resistant bacteria: e.g. **Multidrug-resistant TB**, **Gonorrhea**, **CRE**
  - Healthcare-Associated Infections: e.g. **MRSA**, ***C. difficile***
  - **Pertussis**
  - **Cholera**
  - Anthrax
  - Tularemia

Methicillin-resistant *Staphylococcus aureus*

- Gram +ve cocci
- Normal flora of humans:
  - Nasal membranes/skin
- Opportunistic infections:
  - Mostly skin infections
  - Focal damage
    - 53% purulent wounds
    - 61% abscesses
  - Systemic disease
- Nosocomial infections
- Historically termed MRSA strains are oxacillin and methicillin resistant
- They are also resistant to all b-lactam agents, including cephalosporins & carbapenems





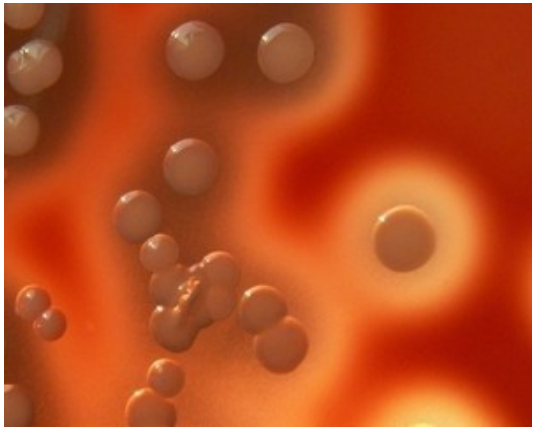


MRSA (Detection)

- Altered penicillin-binding protein (PBP2a) coded by *mec*A gene
- Challenging due to underexpression *in vitro*
- Broth microdilution testing
- Cefoxitin disk screen test
- Latex agglutination test for PBP2a
- Mueller-Hinton agar with 6 µg/ml of oxacillin (supplemented with 4% NaCl)
- PCR for *mec*A gene

MRSA

- All members are highly resistant to drying, extremes in pH, and high temperatures (Biofilm formation)
- Increase in community acquired cases, with infections associated with schools, prisons, & shared athletic equipment
- MRSA was responsible for 94,000 life-threatening infections and nearly 19,000 deaths in the US
- VRSA



S. aureus is β -hemolytic on blood agar

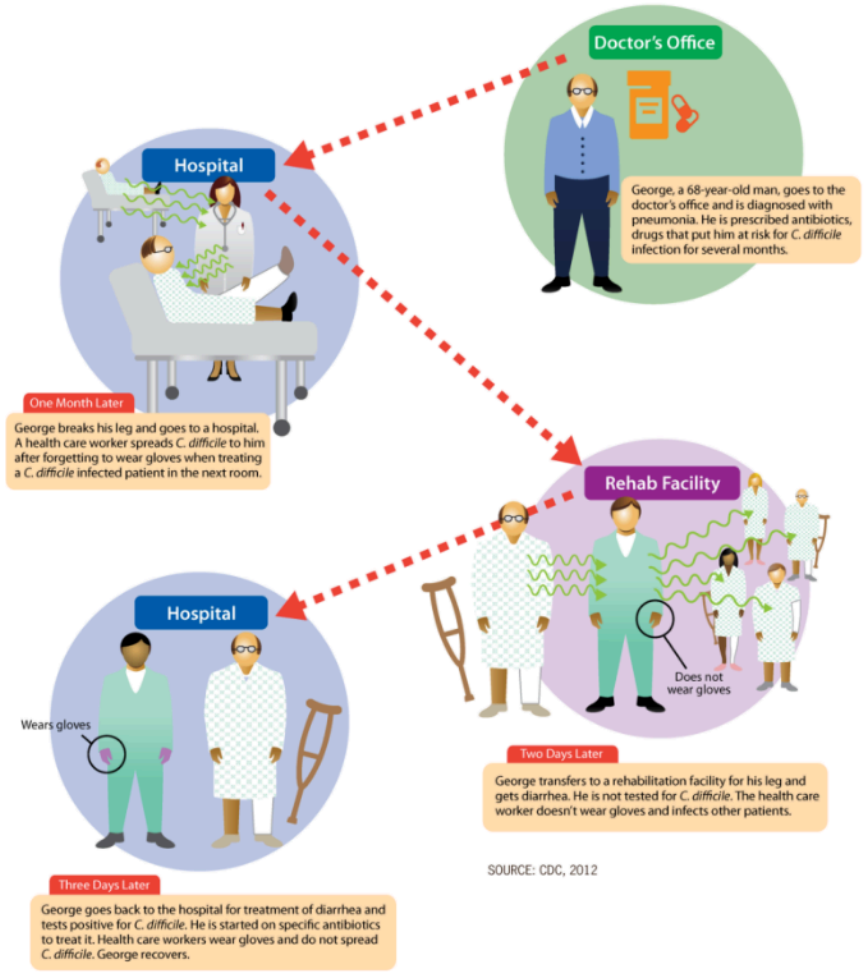
"the thoughtless person playing with penicillin is morally responsible for the death of the man who finally succumbs to infection with the penicillin-resistant organism. I hope this evil can be averted" Sir A. Fleming, 1945

The rise of *C. difficile*

Hypervirulent epidemic strain of CDI (B1/NAP1)

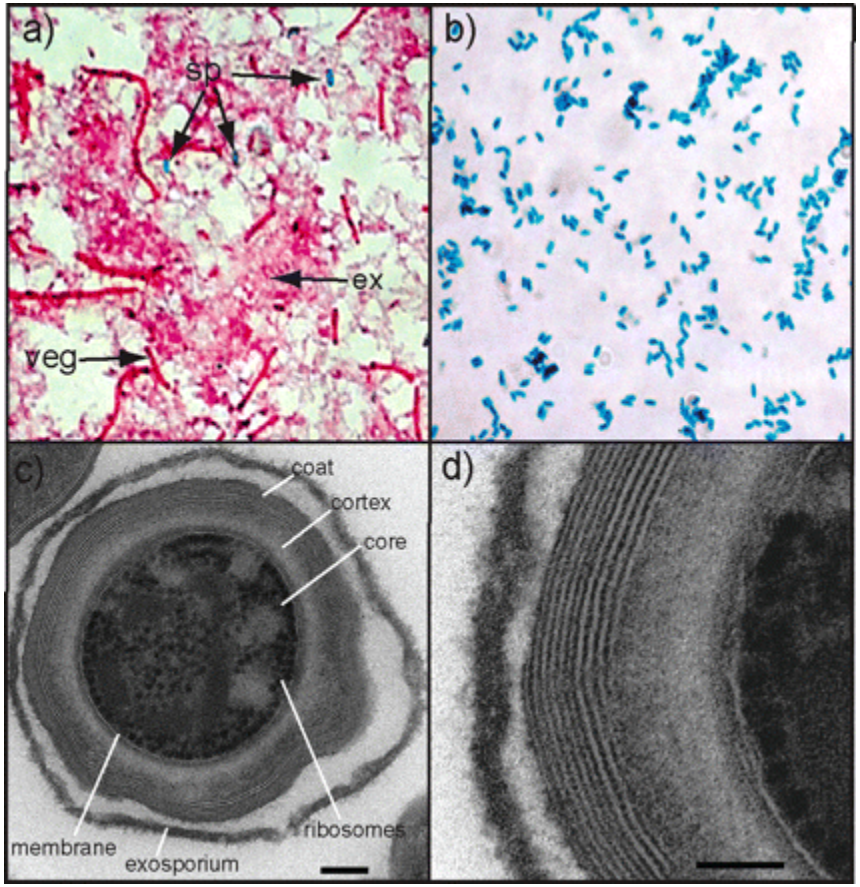
- Implicated in outbreaks throughout the US, Canada & Europe
- Now seen in at least 40 states, exhibit:
  - Greater toxin production
  - Greater antimicrobial resistance
- Control: infection control, environmental cleaning & antibiotic stewardship

How *C. difficile* spreads?



Identification of *C. difficile*

Gram +ve, spore-forming rods





Requires anaerobic culture



Stools positive for Toxin A (enterotoxin) and/or B (cytotoxin)

Multidrug-resistant TB

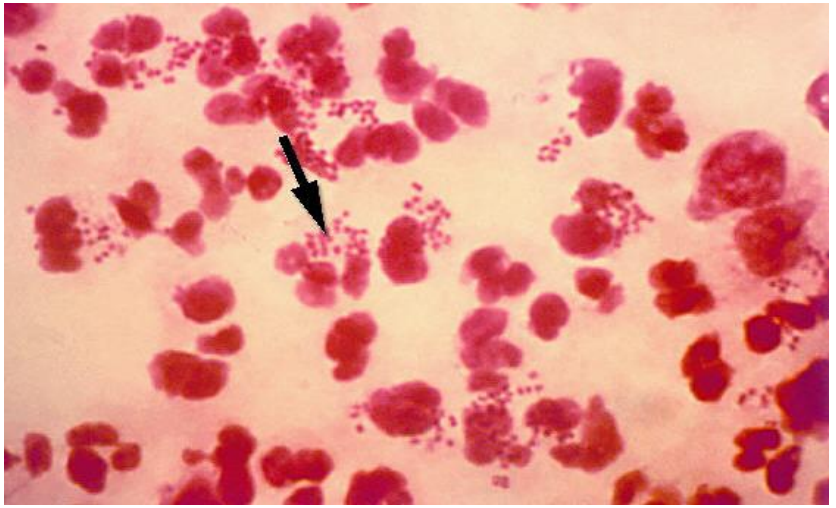
- Prevalent in hospitals, prisons, etc.
- HIV infected patients
- 1 in 106/108 spontaneous mutants resistant to first-line anti-TB drugs
- Combination treatment (*takes longer & require 2nd line drugs*)
- 1st line: INH, RMP, PZA & EMB
- 2nd line: kanamycin, ethionamide, cycloserine, ofloxacin & ciprofloxacin
- Resistance mechanisms identified -> mutations in target proteins
- Multidrug-resistant *M. tuberculosis*: INH & RMP

XDR-TB

- Extensively drug-resistant strains (XDR):
  - Resistant to:
    - INH, RMP, &
    - any fluoroquinolone &
    - at least one 2nd line (amikacin, capreomycin, or kanamycin)
- Globally recognized but prevalence is underestimated due to limited resources
- 64% more likely to die
- 480,000 incident cases of MDR-TB worldwide 9.5% are XDR-TB
- Treatment options include: BPal, BPaLM

Multidrug-resistant Gonorrhea

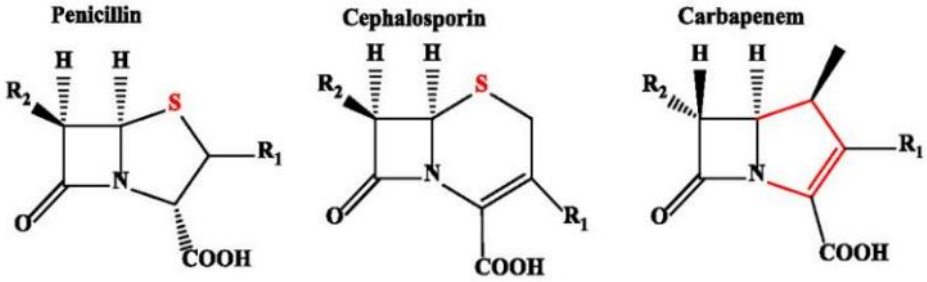
- *Neisseria gonorrheae*
- 70% homology to *N. meningitidis*
- Un-capsulated
- Antigenic variation (pili, Opa, & LOS; 1 in 103)
- Resistance:
  - Beta lactamase (plasmid),
  - Tetracycline-R (chromosomal),
  - Fluoroquinolones-R
  - Cephalosporin-R (Cefixime/Ceftriaxone)
- CDC Guidelines: only one regimen of dual therapy for the treatment of gonorrhea - injectable ceftriaxone + oral azithromycin



http://diseasespictures.com/wp-content/uploads/2012/08/Neisseria

Carbapenem-resistant *Enterobacteriaceae*

- Carbapenems - most reliable last-resort treatment for bacterial infections
- Combination of b-lactam ring & carbapenem nucleus
- *Klebsiella*, *E. coli,* & *Pseudomonas*
- Mechanisms of resistance:
  - Efflux pumps
  - Decrease permeability of outer mem
  - Carbapenemases are the most clini
    - *Klebsiella pneumoniae* carbapene
    - New Delhi Metallo-beta-lactamase (NDM), &
    - Verona Integron-Mediated Metallo-β-lactamase (VIM)
- Most partially inhibited or not inhibited by inhibitors like clavulanic acid
- Nosocomial infections



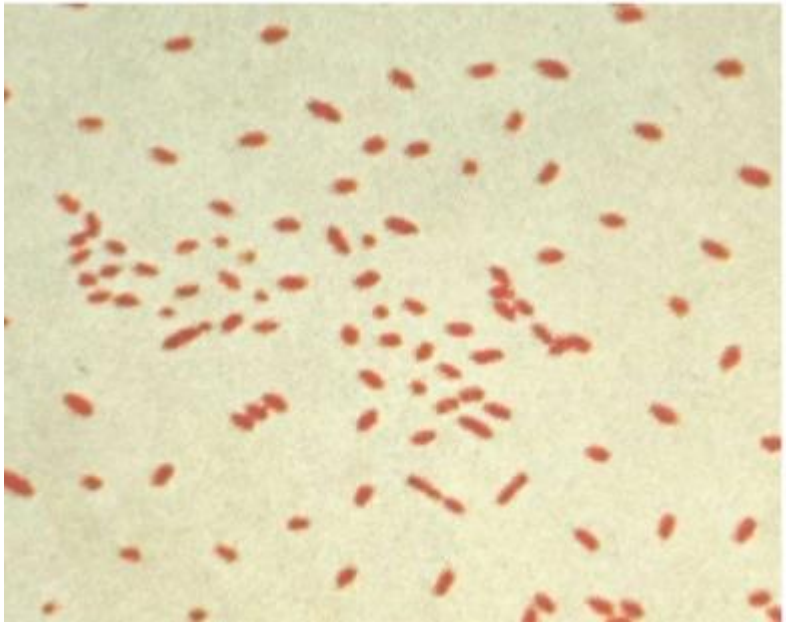
http://aac.asm.org/content/55/11/4943/F2.large.jpg

Carbapenem-resistant *Enterobacteriaceae*

- Some CRE have become resistant to most available antibiotics
- Mortality up to 50%
- Contact precautions, alert facility, infection control & laboratory staff
- Tx:
  - Best: combination therapy with two or more drugs including at least one carbapenem (carbapenem's MIC is ≤ 4 mg/L)
  - Last resort (old drugs): polymyxins (renal toxicity), fosfomycin

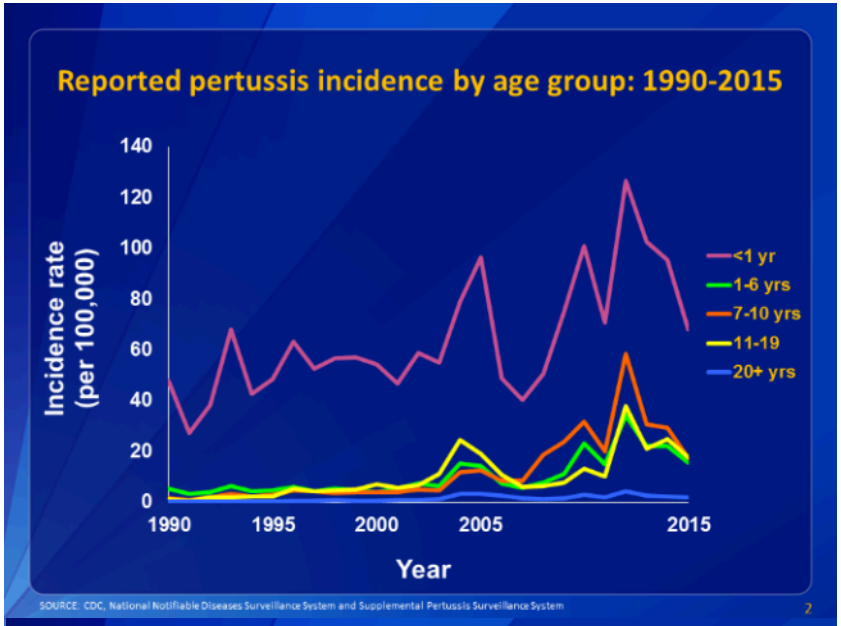
Pertussis

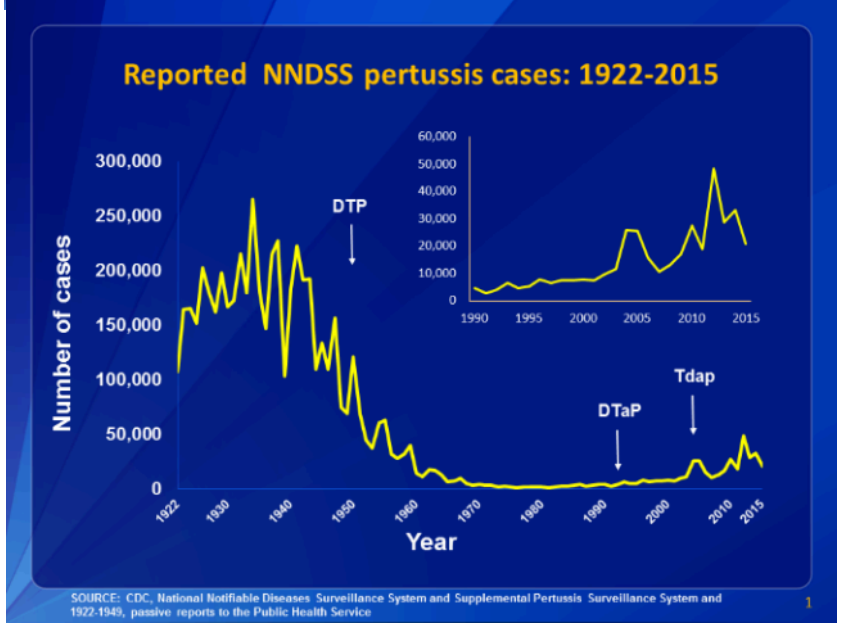
- *Bordetella pertussis*
- Capsulated
- several toxins:
  - Pertussis toxin (A/B)
  - Adenylate cyclase Toxin
  - Dermonecrotic toxin
  - Tracheal cytotoxin
  - Hemolysin
- Stages: catarrhal (infectious), paroxysmal (whoop), convalescent stage
- Immunity is not lifelong



https://image.slidesharecdn.com/bordetella-160415071539/95/bo

Pertussis

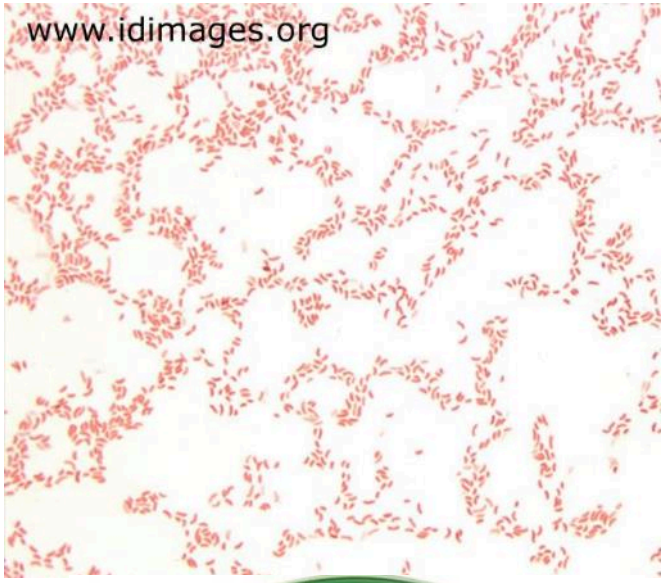


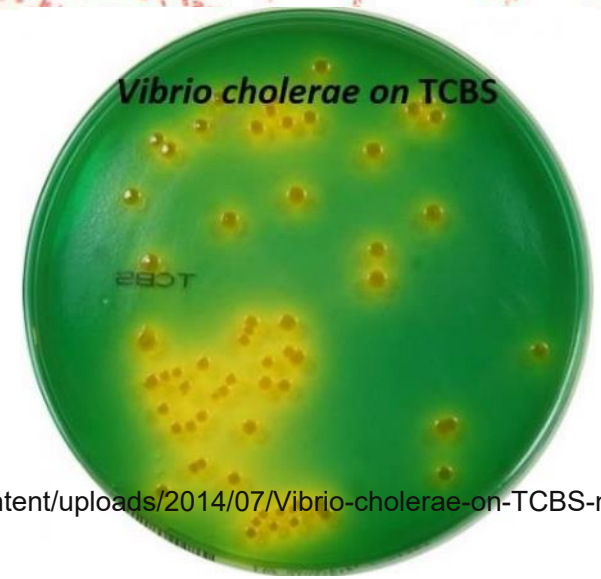


- Endemic worldwide in most densely populated areas
- Highly contagious
- By late 1990s to early 2000 -> rise in cases in US
- Vaccinated children (7-10 yrs.)
  - Waning immunity
  - Antigenic change

Cholera

- Rare in the US (unless there are disasters)
- Marine/surface water
- Alkaliphiles/halotolerant (pH=8.5-9.5)
- Acid-intolerant:
  - 1010 in water
  - 102-104 in food
- Serotype: O1 & O139 (LPS)
- TCBS agar

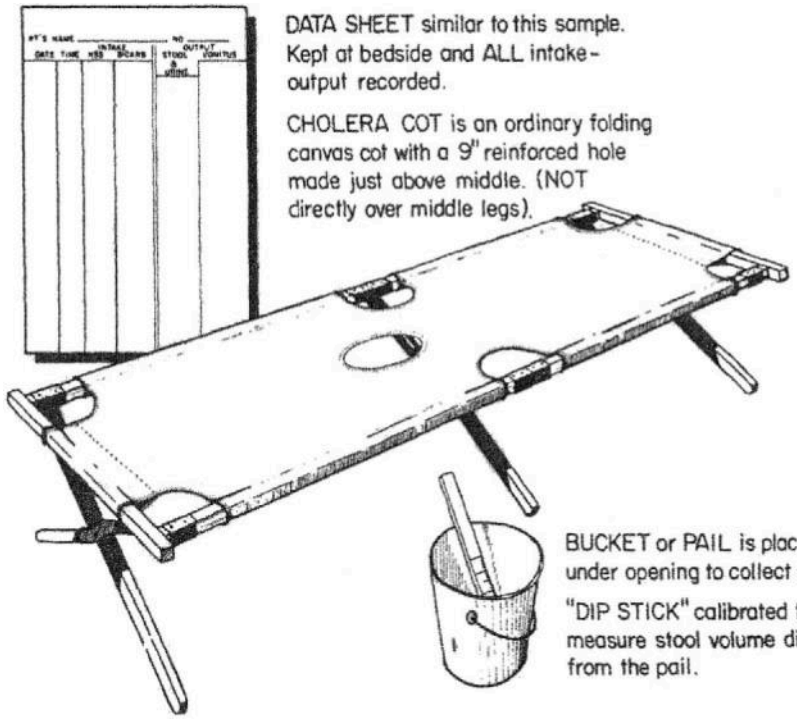




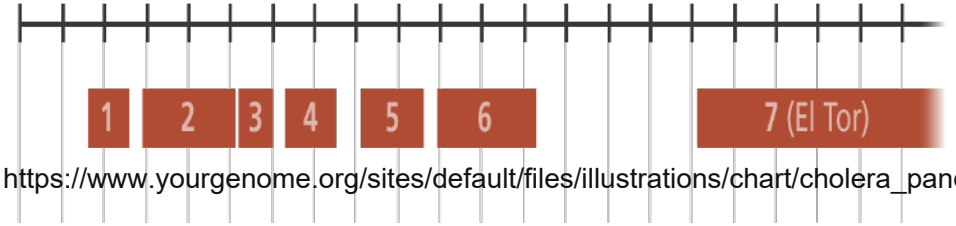
http://www.medical-labs.net/wp-content/uploads/2014/07/Vibrio-cholerae-on-TCBS-me

Cholera

- Non-invasive
- Most infections asymptomatic (~70%)
- Incubation: 0.5-3 days
- Rice-watery stool
- Cholera toxin (A/B)
- Mortality 25-50% (without Tx)
- Tx: electrolyte replacement
- 7 pandemics
- Reportable to CDC
- 2010 Haiti earthquake
- Outbreak in Yemen



http://1.bp.blogspot.com/-r_P_bKtKToE/TmEb7QSSknI/A

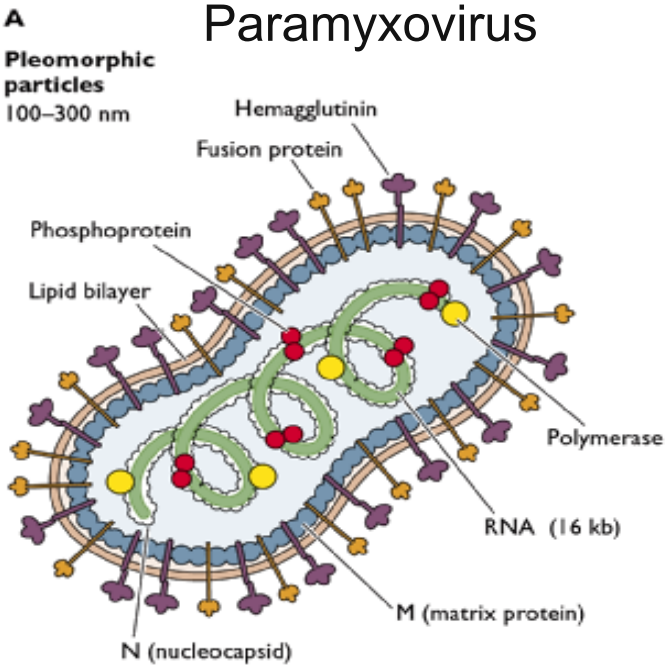


https://www.yourgenome.org/sites/default/files/illustrations/chart/cholera_pander

Measles (Rubeola)

(-) ssRNA
Worldwide
- Leading cause of child death
  - 164,000 deaths in 2008
- Measles Initiative
  - Decrease deaths due to measles
  - American Red Cross, United Nations Foundation, CDC, UNICEF, World Health Organization
  - Began in 2001 because 733,000 worldwide deaths
  - 78% drop in number of deaths by 2008

Paramyxovirus



Measles

U.S.
- Declared eliminated in 2000
- Number of unvaccinated children increases
  - 131 measles cases
  - Increase of 37 in 2004
  - Of the 131
    - 112 not vaccinated
      - 63 not vaccinated due to parents' beliefs

Lancet (1998)
- Linked MMR vaccine to autism and IBD
- Vaccination decreased 92% to <80% in U.K.
- ~125,000 children in U.S. remain unvaccinated

Number of measles cases by year since 2010

Year	Cases
2010	63
2011	220
2012	55
2013	187
2014	667
2015	188
2016*	70
2017**	21

*Cases as of December 31, 2016. Case count is preliminary and subject to change.

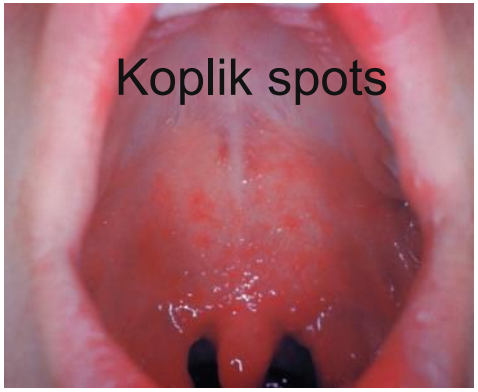
Cases as of February 25, 2017. Case count is preliminary and subject to change. **Data are updated monthly.

Source: Morbidity and Mortality Weekly Report (MMWR), Notifiable Diseases and Mortality Tables

Measles

Spread through respiratory secretions
- Coughs, sneezes
- Virus survives on surfaces up to 2 hours
- Dense rash on face, trunk, arms, legs
- Sore throat, high fever (up to 41°C), cough, runny nose, red eyes, light sensitivity
- Signs/symptoms occur 7–18 days after infection
- **Koplik spots** seen in mouth
  - Red spots with bluish white centers





Measles: Diagnosis & Treatment

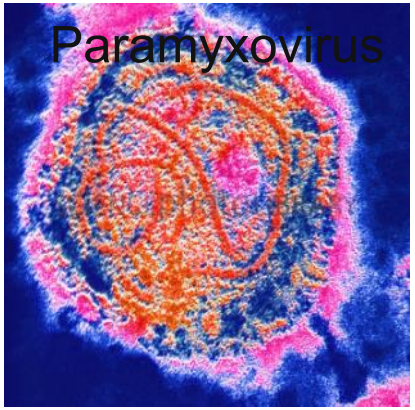
- Diagnosis
  - Test for measles Abs to determine immune status
  - Clinical diagnosis
    - Characteristic rash & fever
  - Only one antigenic strain of measles
    - Several strains can be differentiated genetically
  - Viral culture
    - Nasal or throat swab
  - RT-PCR
- Treatment
  - Acetaminophen or ibuprofen (fever)
  - No tx to decrease the viral load

Mumps

(-) ssRNA

Signs & symptoms

- 15–20% are asymptomatic
- Swollen parotid salivary glands
- Submaxillary, sublingual salivary glands may be enlarged
- Fever, headache, muscle aches, loss of appetite, fatigue





(a)



(b)

Mumps

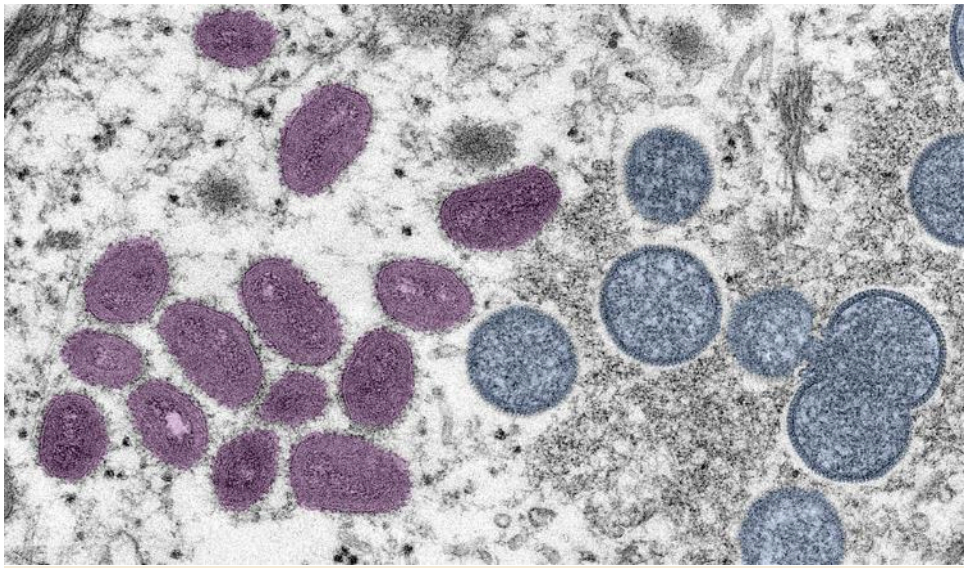
- Spread
  - Direct contact with secretions
  - Indirect by contaminated items (fomites)
- Contagious
  - 1–2 days before swelling; up to 5 days after swelling
- Lab analyses:
  - Viral culture
  - RT-PCR
- Lab to detect IR
  - Anti-mumps IgM
  - 4-fold rise in IgG
- Mumps, rubella, & measles are reportable

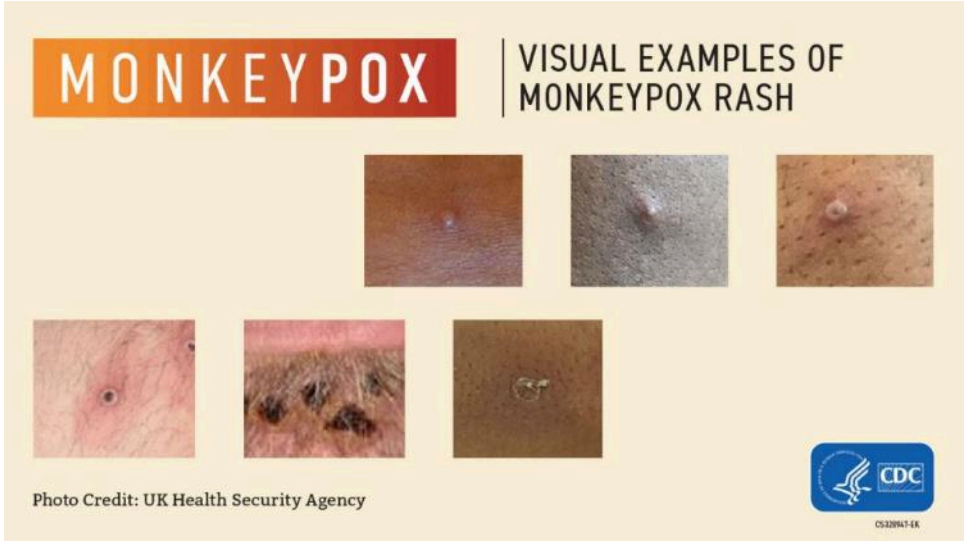
Polio

- Was scheduled by the WHO to be eradicated in 2000
- Then was postponed to 2010
- NY city declared state of emergency
- Some NY counties have 60% vaccination rate (way lower to achieve herd immunity)
- Primarily a GI illness -> poliomyelitis

MonkeyPox

- 1st discovered in 1958
- Endemic to central & western Africa
- Transmission: direct contact
- Tecovirimat (viral release)
- JYNNEOS vaccine (live attenuated non-replicating)



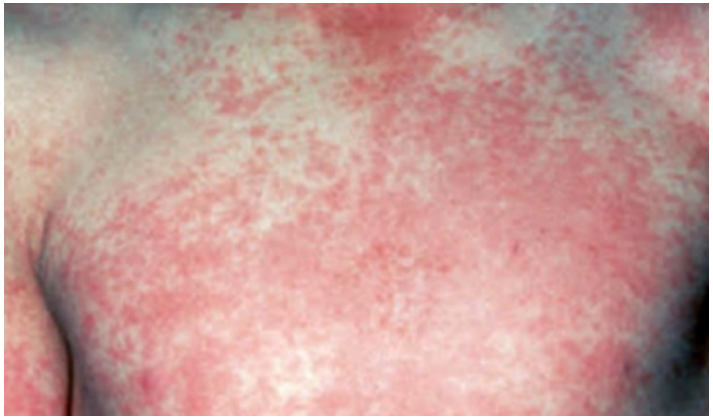


Dengue Fever

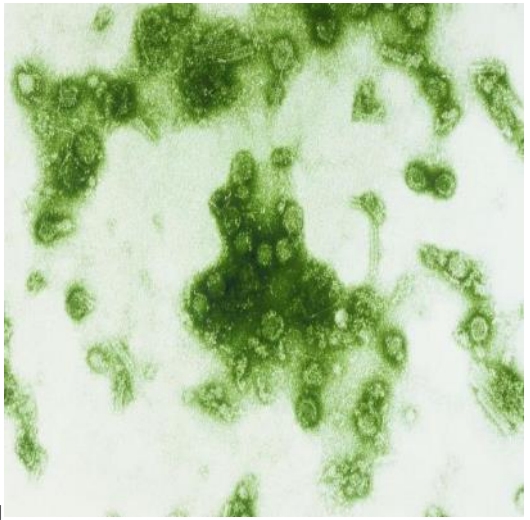
- (+) ssRNA
- Breakbone fever
- *Aedes aegypti*
- Incubation: 4-7 days
- Malaise, chills, headache, fever, & rash
- Pain in back, joints, muscle, bone & eyeballs (viral load)



https://i2.wp.com/offlineclinic.com/wp-content/uploa



http://www.healthmds.org/wp-content/uploads/2017/05/Dengue-fever-rash.png



https://www.insectshield.com/images2/DengueFever_16x

Dengue fever

- DHF occurs in few
- Non-neutralizing Abs
- Vasoactive cytokines -> shock -> death
- Dx: RT-PCR
- 4 serotypes of virus
- No cross-reactive immunity (opposite)

- Endemic (tropics)
- 50 million cases worldwide
- 400k DHF
- DHF mortality: 15% with no Tx; 1% with supportive (no antiviral)
- Key West, FL and TX
- No vaccine yet

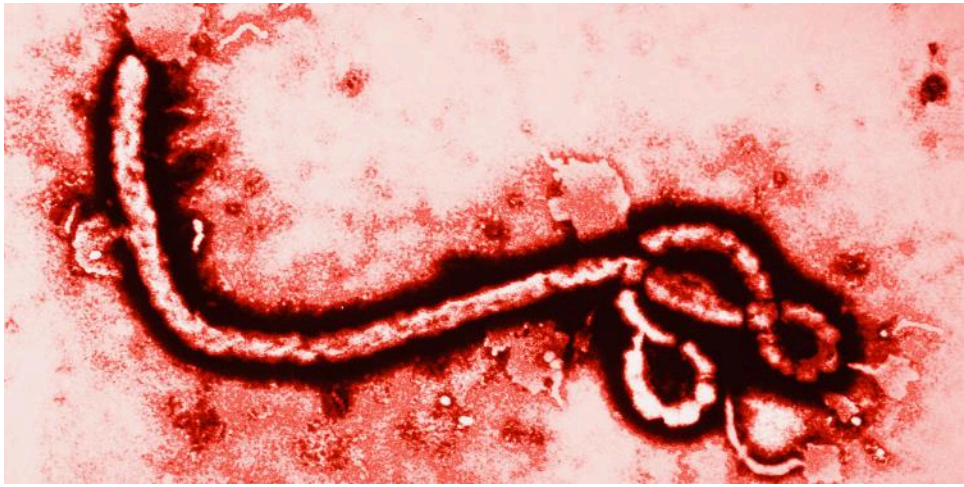




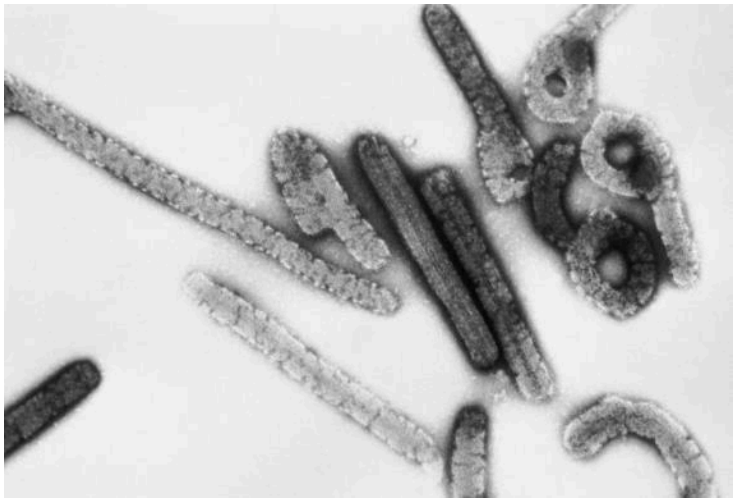
https://www.cdc.gov/dengue/epidemiology/index.html

African hemorrhagic fevers

- Ebola/Marburg
- Filoviridae
- (-) ssRNA
- Reservoir & vector: African fruit bats
- Incubation:
  - Marburg: 3-9 days
  - Ebola: 2-21 days
- Symptoms: headache, fever, sore throat, myalgia then abdominal pain, vomiting, diarrhea, & rash
- Internal/external bleeding -> shock & death



http://i.huffpost.com/gen/2097108/thumbs/o-EBOLA-VIRUS-facebook.jpg



https://upload.wikimedia.org/wikipedia/commons/9/99/Mar

African hemorrhagic fevers

- Ebola (Zaire, Sudan, Reston & Ivory Coast; can differ by 40%)
- 1st recorded cases 1967 (Marburg), 1976 (Ebola)
- Current outbreak of Ebola in Zaire
- Last year outbreak of Marburg in Uganda
- Highly virulent (**BSL 4**)
- Tropic to reticuloendothelial cells (immunosuppressive)
- High titers in most tissues
- Mortality (20-90%)

African hemorrhagic fevers

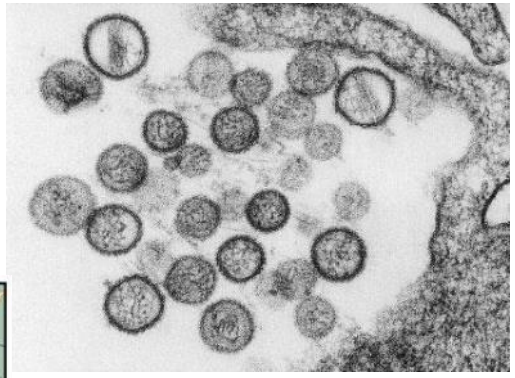
- 2014 - Ebola largest outbreak
  - Guinea, Liberia & Sierra Leone
  - 10,000 death
  - Several imported cases
  - 1 in the US (exposed some flawed control measures)
- Abs protective
- Detection: ELISA & RT-PCR (special care with handling sera)
- Person to person spread (bodily fluids/blood)
- Patients isolated
- Tx: Providing fluids and electrolytes (body salts) through infusion into the vein (intravenously).
  - Offering O2 therapy
  - Using medication to support bp, reduce vomiting & diarrhea
  - Manage fever & pain
  - Treating other infections, if they occur
- No approved antiviral treatment (Regeneron (REGN-EB3) & mAb114) or vaccine (experimental)

Hantavirus Pulmonary Syndrome

- 1st outbreak in 1993 (US)
- Sin Nombre virus
- (-) ssRNA segmented
- Reservoir: deer mouse (prevalence in 10% tested)
- Mortality: 30%
- Symptoms: fever, headache, myalgia -> progressive pulmonary edema -> severe respiratory compromise
- No hemorrhage
- 4-fold rise in IgG
- RT-PCR
- Ribavirin (some benefit)



http://zipcodezoo.com/images/thumb/9/93/Peromyscus_maniculatus_12

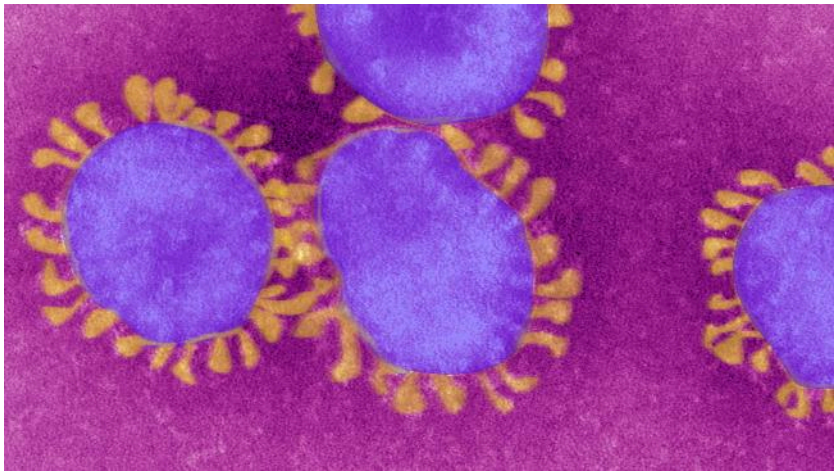


//images.medicinenet.com/images/govern



Coronaviruses

- (+) ssRNA
- Up to 30% of common colds
- Human coronavirus infections occur during the *winter & early spring*
- Novel Coronaviruses:
  - SARS (2003); zoonosis (bats)
  - MERS (2012; 40% mortality)
  - SARS-CoV-2



http://mediad.publicbroadcasting.net/p/shared/npr/201302/17239691

ere acute respiratory syndrome (SARS)

- Outbreak: November 2002, highly contagious & severe atypical pneumonia were observed in the Guangdong Province of southern China
- Virus spread to Hong Kong in February 2003 by a physician who died 10 days after admission into local hospital -> a mysterious death
- Similar outbreaks occurred at different local communities subsequently
- The virus was identified by Hong Kong, the U.S. & Germany in March 2003
  - >8,450 cases, 810 death, 30% healthcare workers, 32 countries



Natural hosts of SARS

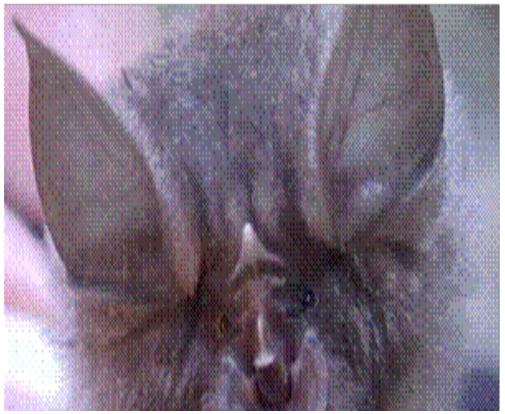
SARS may have originated in wild horseshoe bats in China (reservoir)

Indications:

- 92% similar to the human SARS virus from 3 species of bats found at the Zoology at the Chinese Academy of Sciences, Beijing
- Infected bats shows little or no symptoms

Direct human infection is not likely to occur

- The differences is in the genes which able the binding of virus particle to host cell
- Virus may jump to an intermediate animal such as the civet cat & adapt to infect humans



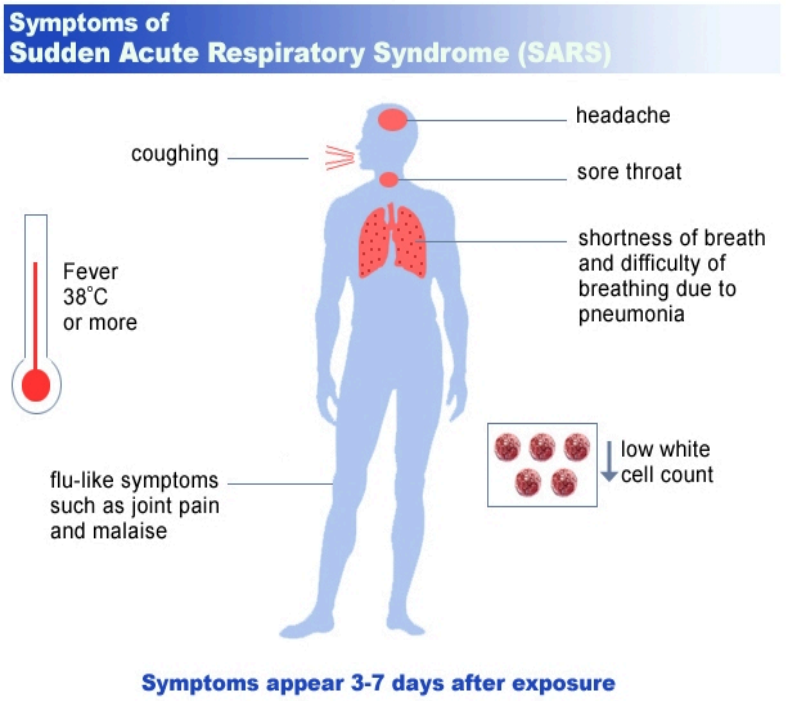


Transmission of SARS

- Close contact with large respiratory droplets
- Transmission by fomites possible
- Those severely ill more infectious (attack rate of >50% in some hospital staff)
- Infectivity increases during 2nd week of illness
- May remain infectious up to 10 days once afebrile

SARS: Symptoms

- Clinical course:
  - Short incubation period (2-3 days)
  - Intensive care if needed occurs about 2 weeks to a month after onset of symptoms
- There are generally 3 phases:
  - Week 1: cold-like symptoms, fever, myalgia, chill and a sore throat
  - Week 2, recurrence of fever. Onset of diarrhea, and oxygen desaturation
  - Only 20% of patients reach this phase, requires ventilatory support
  - The fatality rate is about 10% upon infection (average but burden higher for populations over 65 years old)
  - This information in unvaccinated/boosted



SARS: Diagnosis

- RT-PCR positive for SARS virion using validated methods on at least 2 different clinical specimens
- ELISA Antigen test
- Seroconversion (gold standard)

(negative antibody test on acute specimen followed by positive test on convalescent sera or > 4 rise in antibody titre between acute and convalescent sera)

Treatment for SARS

- No specific antiviral drugs available
- Vaccines are under development:
  - In China, a second-phase human trials of a SARS vaccine (inactivated)
    - Have safety concerns
  - The first U.S. SARS vaccine trial began at the NIH by using recombinant plasmid for S protein expression
  - Development of transgenic tomato & tobacco expressing SARS virion S protein
  - An experimental attenuated Vesicular Stomatitis Virus (VSV)-based vaccine

Prevention of SARS



Wearing N95 mask was one of the main public preventive measures against SARS

- Close contacts of a probable or suspect case of SARS are at high risk of becoming a case (should be isolated)
- Travelers from areas with recent local transmission of SARS should be given advice about whom to contact, and what to do if they develop a fever within their potential incubation period
- Health departments should assign staff to undertake contact tracing and daily review of contacts in the community.
- Recovered patients quarantined at home for at least 14 days after discharged from hospital

SARS-CoV-2

- (+) ssRNA enveloped
- Novel virus but not unique
- Source:
  - Reservoir: bats
  - Intermediate host: pangolin (99% match)
- Very similar to SARS-CoV-1 (more infectious and lower mortality
- Tx: Supportive, dexamethasone, remdesivir, Paxlovid and molnupiravir
- Evushield high affinity Ab cocktail

SARS-CoV-2

- Prevention:
  - Mask (N95 or surgical; ***cloth masks are NOT recommended due to poor efficiency***)
  - Social distancing effective
  - Avoid indoor & poor ventilated settings (restaurants, bars, places of worship, gyms)
  - Vaccine

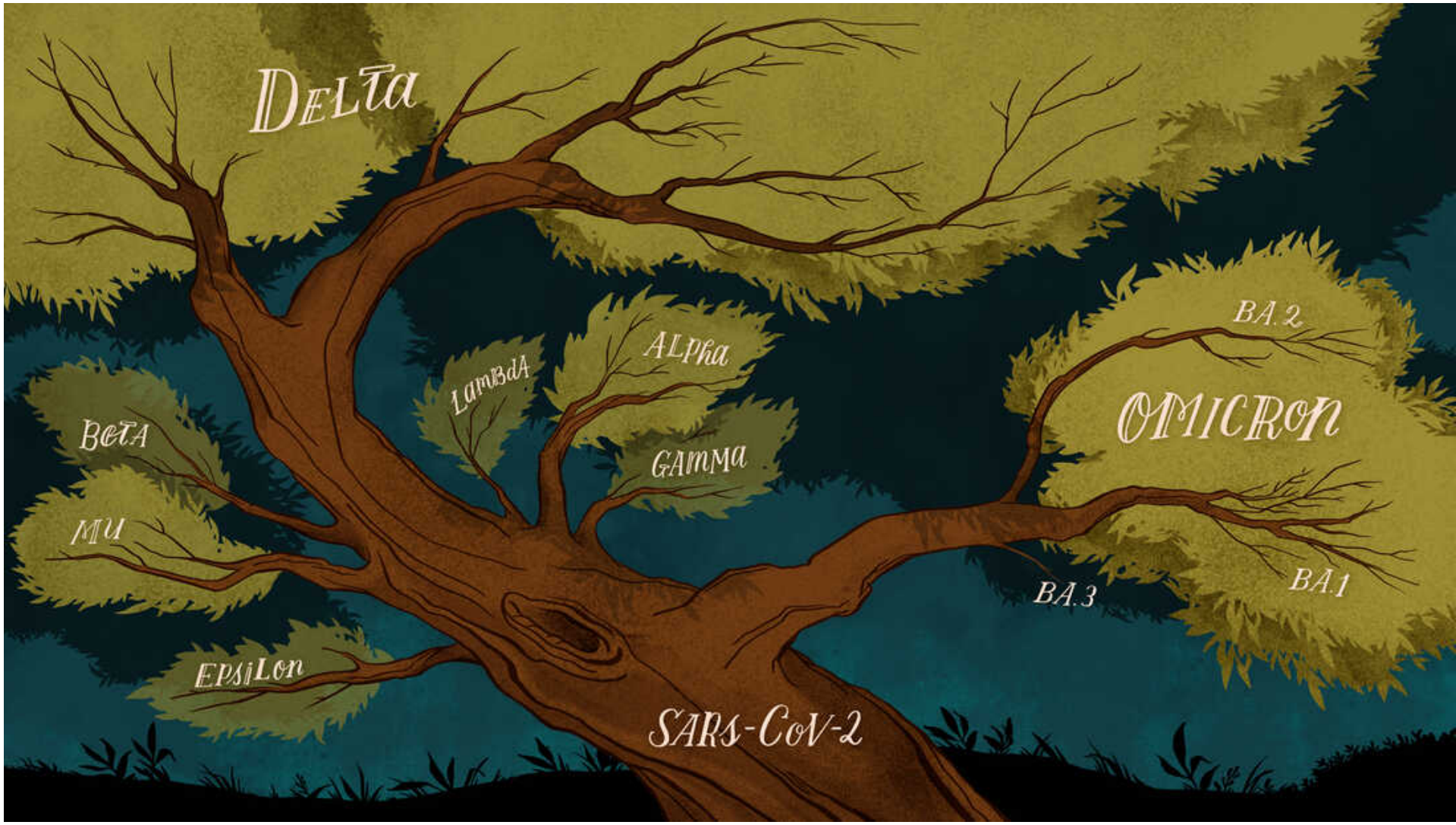
Transmission of SARS

- Close contact with large respiratory droplets or airborne
- Transmission by fomites possible but rare
- Infectivity starts during asymptomatic phase and well into 2nd week of illness or if person is shedding virus
- May remain infectious up to 10 days once afebrile
- *Differs between different variants*

Treatment for SARS

- Paxlovid and molnupiravir
- Vaccines:
  - Pfizer-BioNTech and Moderna mRNA vaccines
    - Primary and updated omicron dose
  - Johnson & Johnson's Janssen, Oxford/AstraZeneca, and Sputnik viral vectors
  - Novavax subunit vaccines
  - Inactivated vaccine in China (Sinovac & Sinopharm) and India (Covaxin)
  - New inhaled viral vector vaccines developed by China (CansinoBio) and India (Baharat)

Evolution of SARS



Variants

- Omicron
  - Predominant strain in US
  - Original strain (BA.1) is no longer circulating
  - First discovered in Botswana
  - More infectious than delta due to mutations on spike protein
- Delta
  - India, late 2020
  - 80 to 90% more transmissible than the Alpha variant
  - more likely to result in hospitalization in the unvaccinated

Variants

- Beta
  - South Africa, Fall 2020
  - CDC: “Beta was about 50% more contagious than the original coronavirus strain”
  - Pfizer-BioNTech, Moderna, and Johnson & Johnson also reported less protection against Beta
- Alpha
  - First of the variants
  - England, November 2020
  - Subsided with more aggressive delta variant
  - more likely to result in hospitalization in the unvaccinated

Influenza viruses

- Orthomyxovirus
- -ssRNA (segmented)
- Spike proteins:
  - H: binds to host cell membane
  - N: cleaves sialic acid for virus exit from the cell
- Types A, B & C
- Multiple hosts

H - hemagglutinin

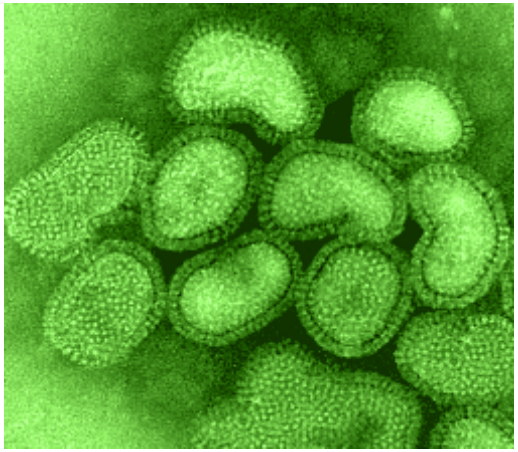
N - neuraminidase

helical nucleocapsid (RNA)

lipid bilayer membrane

polymerase

M1 protein



Influenza virus

- Antigenic drift
  - Spontaneous mutations in viral genome
- Antigenic shift
  - Re-assortment in doubly infected cell in which a virion is packed with a segment of RNA from a different virus, creating a new virus

Pandemics

1890 **H2N8**
1900 **H3N8**
1918 **H1N1** “Spanish Influenza” 20-40 million deaths
1957 **H2N2** “Asian Flu” 1-2 million deaths
1968 **H3N2** “Hong Kong Flu” 700,000 deaths
2009 **H1N1** Re-emergence
between 18,000 US death (151,700-575,400 people worldwide

1918 Spanish flu pandemic



The influenza epidemic taxed the resources of The Red Cross.



Military physicians were baffled by the mysterious illness that was striking young, healthy soldiers.



A nationwide casket shortage was evidence of a mounting death toll.

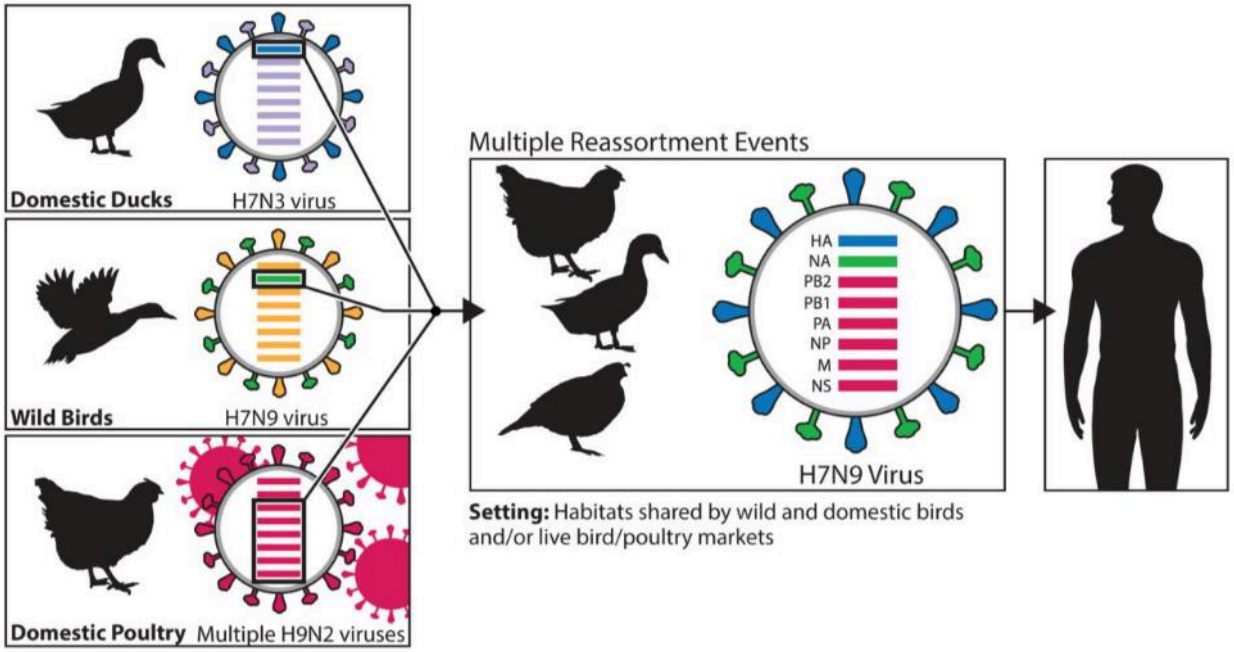


As sailors and soldiers fell ill, doctors puzzled over the mystery illness they were confronting.

Highest mortality in people 20-40 years of age - 675,000 Americans died of influenza - 43,000 U.S. soldiers died of influenza

Avian Strains

- H5N1 (China, 1997)
- H7N9 (China, 2013)



The eight genes of the H7N9 virus are closely related to avian influenza viruses found in domestic ducks, wild birds and domestic poultry in Asia. The virus likely emerged from "reassortment," a process in which two or more influenza viruses co-infect a single host and exchange genes. This can result in the creation of a new influenza virus. Experts think multiple reassortment events led to the creation of the H7N9 virus. These events may have occurred in habitats shared by wild and domestic birds and/or in live bird/poultry markets, where different species of birds are bought and sold for food. As the above diagram shows, the H7N9 virus likely obtained its HA (hemagglutinin) gene from domestic ducks, its NA (neuraminidase) gene from wild birds, and its six remaining genes from multiple related H9N2 influenza viruses in domestic poultry.

Challenges?

- No treatment or last resort treatments
- Rare
- Not-equipped
- Monitoring emerging antimicrobial resistance
- Decline in antimicrobial susceptibility testing (proliferation Nucleic amplification testing bypass culture)
- Pandrug resistance (PDR)

Summary

- Defined emerging infectious disease
- Causes that lead to the emergence of these diseases and efforts to combat that.
- Viral and bacterial pathogens, their properties, pathogenesis, geographical distribution, etc.
- Challenges emerging infectious diseases present to the medical community.

References

- Engleberg NC et al. 2022. **Schaechter's Mechanisms of Microbial Disease, 6**th Ed. Lippincott Williams & Wilkins Publishers.
- K. C. Carroll, et. al. 2016. **Jawetz, Melnick, & Adelberg's Medical Microbiology**, 27ed. McGraw-Hill Education.
- Tao, LE, V. Bhushan and M. Sochat. 2026. **First Aid for the USMLE Step 1**; McGraw Hill Publication.

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